$\mathbf{J}$
$\times\mathbf{H}$
$\mathbf{E}$
$\mathbf{H}$
z
$\mathbf{J}$
$\mathbf{S}$
$\mathbf{L}$
(r, ϕ, z)
$e^{i\ell\phi}$
$z =$
$\ell\hbar$
L_z
ℓ
$\hbar$
$\partial E_x/\partial x$
$\partial E_y/\partial y$
$\nabla\cdot$
$\mathbf{E} =$
$\frac{\partial E_x}{\partial x} +$
$\frac{\partial E_y}{\partial y} +$
$\frac{\partial E_z}{\partial z} =$
0
E_z
E_x, E_y
$s =$
± 1
$s =$
$+1$
$_{in} =$
$\mathbf{E}_0(r, z)(\hat{x}+$
$is\hat{y})e^{ikz}$
$_{in} \propto$
$(\hat{x}+$
$is\hat{y}) =$
$(\cos\phi\hat{\rho}-$
$\sin\phi\hat{\phi})+$
$is(\sin\phi\hat{\rho}+$
$\cos\phi\hat{\phi})$
θ
$\hat{\phi}$
z
$\hat{\rho}$
z
$\hat{\rho}$
$\propto$
$\cos\phi+$
$is\sin\phi$
$\cos\phi+$
$is\sin\phi =$
$e^{is\phi}$
θ
ϕ_p
E_z
$_z \propto$
$\int_0^{2\pi}(\cos\phi_p+$
$is\sin\phi_p)e^{ik\rho\cos(\phi_p-\phi)}d\phi_p$
$\int_0^{2\pi} e^{is\phi_p}e^{ik\rho\cos(\phi_p-\phi)}d\phi_p =$
$e^{is\phi}\int_0^{2\pi} e^{is\alpha}e^{ik\rho\cos\alpha}d\alpha$
$e^{is\phi}$
$J_1(k\rho\sin\theta)$
$z =$
$0 =$
$_z =$
$(\rho, \phi, z) =$
$-iAe^{is\phi}I_1(\rho, z)$
A
$I_1(\rho, z)$
$NA =$
$n\sin\alpha$
$_1(\rho, z) =$
$\int_0^{\alpha}\cos^{1/2}\theta\sin^2\theta P(\theta)J_1(k\rho\sin\theta)e^{ikz\cos\theta}d\theta$
θ
α
$P(\theta)$
J_1
E_z
$e^{is\phi}$
ℓ
$s \equiv$
$+1$
$s =$
$+1$
$\ell =$
$+1$
z
$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\phi^2} + 2ik\frac{\partial}{\partial z}\right)u(r, \phi, z) =$
0
$\exp(i\ell\phi)$
2π
ℓ
$\ell =$
$0, \pm 1, \pm 2$